

256-QAM Communication System Simulation

Simulating Digital Communication Systems with MATLAB/Simulink

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Abstract—This paper analyzes the performance of 256-QAM, QAM with OFDM, and M-PSK modulation schemes under additive white Gaussian noise (AWGN) and phase noise. Simulations in MATLAB/Simulink demonstrate that QAM outperforms PSK due to its larger Euclidean distance, while OFDM introduces slight BER degradation due to cyclic prefix overhead. Constellation diagrams and BER curves quantify the impact of phase noise and validate theoretical expectations.

I. INTRODUCTION

The lab focused on understanding criticality of designing robust communication systems that can operate effectively in real-world noisy environments. We were to design a communication system with a 256-QAM scheme with AGWN and phase noise to emulate noisy environments. Further exploration on different modulation schemes like M-PSK was also advised while also checking how vary SNR na d phase noise altered the overall signal integrity.

II. BACKGROUND INFORMATION

A. 256-QAM Modulation Scheme

It is a modulation scheme that conveys data by varying both the amplitude and phase of a carrier signal. In this scheme, 8 bits of information are grouped together and mapped to one of 256 distinct symbols in the complex plane. Each symbol corresponds to a unique point in the constellation diagram, where the horizontal axis represents the in-phase component and the vertical axis represents the quadrature component. The advantage of this is the ability to transmit a large amount of data per symbol increasing the overall data rate. However, the dense packing of symbols in 256-QAM also makes the system more sensitive to noise and interference.

B. M-PSK Modulation Scheme

Here, information is conveyed solely through variations in the phase of a constant-amplitude carrier signal. The constellation consists of points uniformly distributed around a circle in the complex plane. Each point corresponds to a unique phase angle, and the number of points (M) determines the order of the modulation (e.g., 8-PSK, 16-PSK).

C. Impairments

AWGN is a common noise model that represents random fluctuations affecting both amplitude and phase, degrading the clarity of the received signal. **Phase Noise** on the other hand is a specific impairment that distorts the phase of the signal. For QAM, phase noise causes rotation and spreading of the constellation points, whereas in MPSK, it leads to errors by shifting the phase angles away from their ideal positions.

III. THE 256-QAM COMMUNICATION SYSTEM

The simulated communication system in this lab is designed to emulate a real-world digital transmission scenario using high-order modulation. The system encompasses several key stages—from data generation and modulation to transmission over an impaired channel, demodulation, and performance evaluation. Below is a detailed discussion of each component and its role in the overall system:

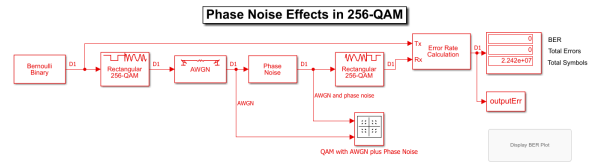


Fig. 1. 256-QAM Comm System

A. Bernoulli Binary Generator

This block simulates the digital data source that would typically be present in a communication system. The randomness of the generated data ensures that the simulation reflects a realistic transmission scenario where the input is unpredictable.

B. Rectangular QAM Modulator

The generated binary data is then passed to a Rectangular QAM Modulator Baseband. Here, the 256-QAM modulation scheme is employed, where each 8-bit group is mapped onto one of 256 distinct symbols in the complex plane.

C. Noise Channels

The **AWGN Channel** block models the impact of background noise that is ubiquitous in real-world environments. By adding white Gaussian noise to the modulated signal, the block simulates random fluctuations in both the amplitude and phase of the signal. The **Phase Noise** block specifically introduces disturbances in the phase of the signal. This simulates imperfections such as oscillator instability or other phase-related fluctuations. Phase noise is particularly detrimental in systems like 256-QAM, where the precise phase of each symbol is crucial for accurate demodulation.

D. Demodulation and Error Analysis

On the receiver side, the noisy signal is processed by a Rectangular **QAM Demodulator** Baseband block, which attempts to map the received, distorted symbols back to their original binary form. To evaluate the performance of the system, an Error Rate Calculation block compares the demodulated output with the original transmitted data, computing the Bit Error Rate (BER). The BER serves as a quantitative measure of how effectively the system can recover the original data in the presence of channel impairments.

E. Constellation Diagrams and BER Plot

Constellation Diagram: This block displays the received symbols in the complex plane, allowing for visual inspection of how AWGN and phase noise affect the ideal constellation. The diagram typically shows a scattering of points around the ideal grid locations, providing immediate insight into the degree of signal degradation.

BER Plot: A callback function is available to generate a plot of BER versus E_b/N_0 for various phase noise levels. This plot illustrates the trade-offs between achieving high data throughput (using a dense constellation like 256-QAM) and maintaining signal integrity in noisy environments.

IV. RESULTS

The first communication system had the 256-QAM Modulator and the two noise channels, AWGN and Phase noise. I first set the SNR (E_b/N_0) to 35 and phase noise at -76. These are ideal conditions in a typical communication system. BER was 0.0001031, Total Errors was 101 and Total Symbols were 9.8e+05

A. Ideal 256-QAM Model with AWGN and Phase Noise

From the constellation diagram shown in fig 2, the received symbols can be seen deviated from their ideal positions. AWGN causes a random scattering effect, where the points are spread out around their intended locations. This scattering increases the chance that a symbol is misinterpreted during demodulation. Phase noise, on the other hand, primarily affects the angular position of the symbols. Its presence results in a rotation or radial spread of the constellation points, making the constellation appear blurred and less distinct. The combined effect of these noise sources in the constellation diagram visually confirms the degradation of signal quality and provides an

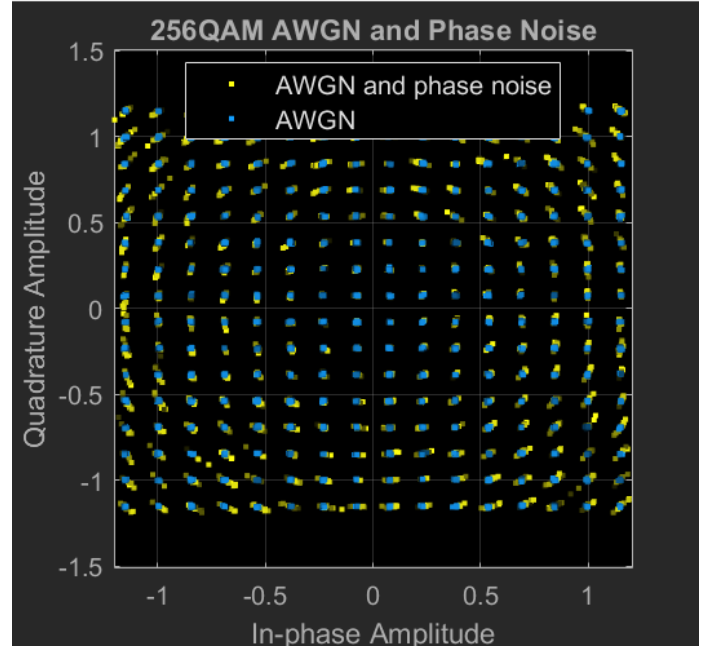


Fig. 2. Constellation Diagram

immediate understanding of how noise disturbs the modulated signal. The BER plot in fig 3 quantitatively evaluates the

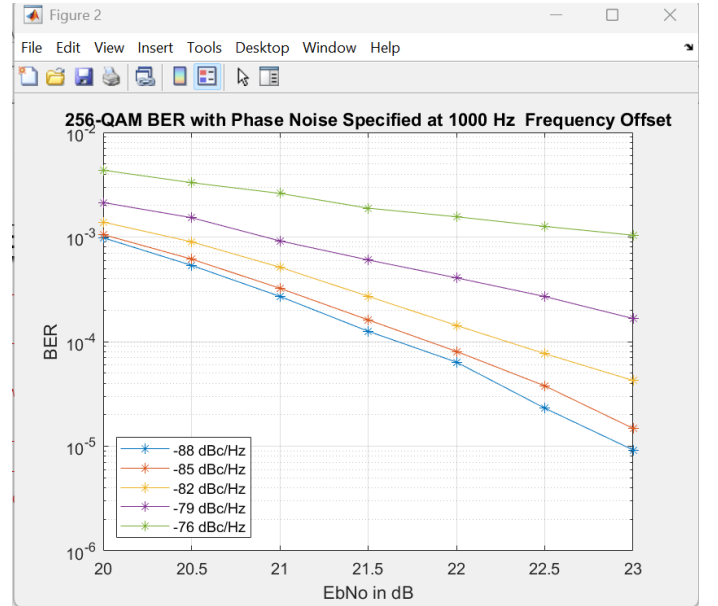


Fig. 3. BER Plot

performance of the communication system by showing the relationship between the bit error rate and the energy per bit to noise density ratio (E_b/N_0). This plot is generated by comparing the transmitted bits with the demodulated bits after the signal has passed through the noisy channel.

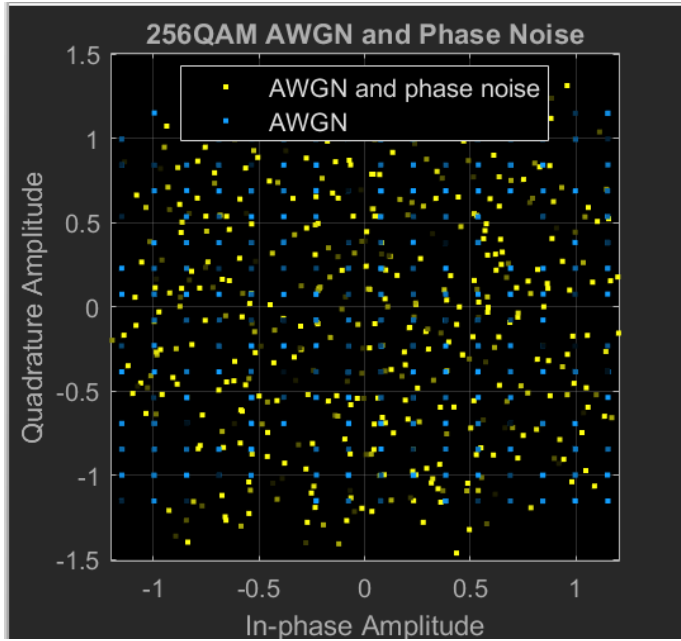


Fig. 4. Constellation Diagram

B. Communication Model with Increased Noise

Fig 4 shows the constellation diagram of the same communication system but with increased AGWN and Phase noise. In this case the symbols are no longer tightly clustered around their ideal positions. Instead, each point spreads out in a random fashion. This scattering makes it increasingly difficult for the receiver to correctly identify the transmitted symbols. SNR was set at 50 and phase noise at -50. There was an overall increase in BER(0.3104) and Total Errors(1201) and a reduction in total symbols(4000).

C. Ideal M-PSK Model with AWGN and Phase Noise

With similar values of AWGN and Phase Noise as the ideal 256-QAM model the simulation produced this result: The constellation diagram consists of M evenly spaced points

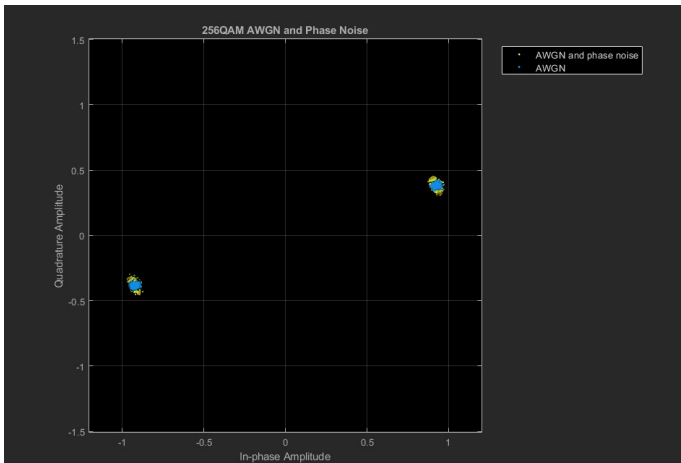


Fig. 5. Constellation Diagram

positioned on a circle in the complex plane. Each point represents a distinct symbol, and the modulation scheme encodes information in the phase of the signal while keeping the amplitude constant. Higher values of M (e.g., 16-PSK, 32-PSK) allow more bits per symbol but make the system more susceptible to noise.

D. M-PSK Model with increased Noise

With increase in noise the constellation diagram now consists of M evenly spaced points on a circle in the complex plane. Each point representing a distinct phase state, while the amplitude remains constant.

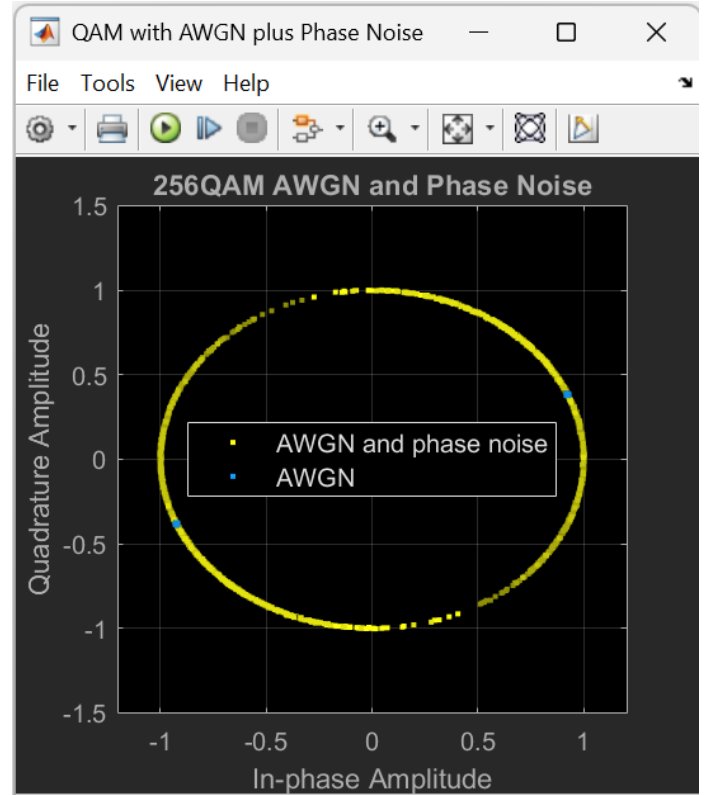


Fig. 6. Constellation Diagram

V. DISCUSSION

This experiment focused on analyzing the performance of a 256-QAM communication system under different channel conditions, initially in an ideal scenario and later under the influence of noise. The system was designed in Simulink, incorporating a binary random generator as the data source, a 256-QAM modulator, a noisy channel with AWGN and phase noise, a demodulator, and performance evaluation tools such as the constellation diagram, BER plot, and error rate calculation blocks. The initial phase of the experiment involved observing the system's performance in an ideal setting, where the transmitted symbols were received with minimal distortion. The constellation diagram displayed a well-structured 16x16 grid, with each point corresponding to a unique 8-bit symbol. The BER was close to zero, indicating that the system functioned

efficiently in the absence of significant noise. This confirmed that 256-QAM is highly effective in high-SNR environments, offering high spectral efficiency.

However, when noise was introduced into the system by decreasing the SNR and increasing the phase noise, significant changes were observed. The constellation diagram exhibited increasing distortion, with points spreading further from their ideal locations due to the impact of AWGN and phase noise. Phase noise, in particular, caused rotational shifts in the symbols, making it harder for the receiver to accurately detect them. As a result, the BER increased, and the number of total errors rose sharply, indicating a higher likelihood of symbol misinterpretation. The BER plot further confirmed this deterioration in system performance, with a noticeable rise in error rates as noise levels increased. This demonstrated that while 256-QAM is efficient under favorable conditions, it is highly susceptible to noise, especially phase noise, due to its reliance on both amplitude and phase for symbol encoding.

To further analyze the effects of noise on different modulation schemes, the experiment was repeated using M-PSK instead of 256-QAM. In M-PSK, symbols are positioned evenly around a circle, making the modulation more resistant to amplitude variations but more susceptible to phase distortions. The constellation diagram of M-PSK under noise showed less amplitude scattering compared to 256-QAM, but with increased rotational shifts due to phase noise. As the noise increased, the BER also rose, but in some cases, M-PSK demonstrated better resilience to amplitude noise than 256-QAM.

This comparison between 256-QAM and M-PSK highlighted the trade-offs between spectral efficiency and noise robustness. While 256-QAM provides higher data rates, it requires a higher SNR to maintain accuracy, making it less suitable for highly noisy environments. M-PSK, on the other hand, is more resistant to amplitude noise, making it preferable in certain scenarios, but it suffers from phase noise at higher modulation orders. The results of this experiment emphasize the importance of adaptive modulation in practical communication systems, where the modulation scheme can be dynamically adjusted based on the channel conditions to optimize both reliability and data throughput.

CONCLUSION

This experiment helped us understand the behavior of a 256-QAM communication system under different noise conditions and compared its performance with M-PSK modulation. It also reinforced key principles in digital modulation and noise analysis, providing a deeper understanding of how different modulation schemes react to real-world impairments. Such knowledge is essential for designing efficient communication systems in practical applications, from wireless networks to satellite communications, where noise mitigation and spectral efficiency are critical considerations.

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